

- 18 The total energy supplied to an electric motor is E . Energy Q is wasted and the remaining energy does useful work.

What is the efficiency of the motor?

- A $\frac{Q}{E}$ B $\left(\frac{Q}{E}\right) - 1$ C $1 - \left(\frac{Q}{E}\right)$ D $\frac{(1-Q)}{E}$

- 19 An object travels between two points. The change in gravitational potential energy ΔE_p of the object is given by

$$\Delta E_p = mg\Delta h$$

where m is the mass of the object, Δh is its change in height and g is the acceleration of free fall.

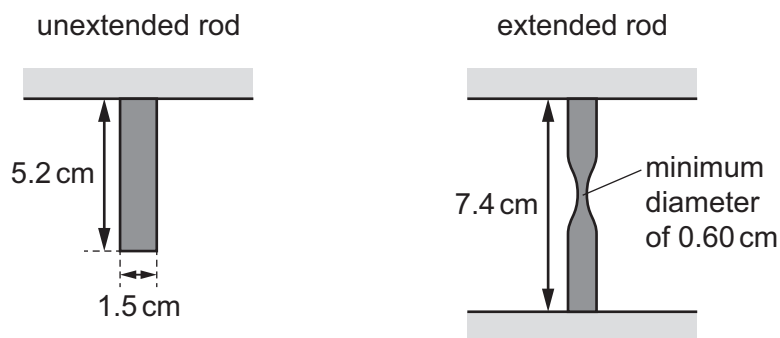
What is a necessary condition in order for the above equation to be valid?

- A The object must have a constant acceleration of 9.81 m s^{-2} .
 B The object must be travelling in a uniform gravitational field.
 C The object must be travelling only in a vertical direction.
 D The resultant force on the object must be equal to its weight.
- 20 A cylindrical steel rod with negligible weight has a diameter of 1.5 cm and a length of 5.2 cm.

The rod is firmly attached at the top end.

The rod is firmly attached to a machine at the other end that applies a constant force of 363 N to the rod.

This causes the rod to extend to a length of 7.4 cm and to have a minimum diameter of 0.60 cm in the position shown.



What is the maximum stress acting on the steel rod when the length is 7.4 cm?

- A $2.1 \times 10^6 \text{ Pa}$ B $3.2 \times 10^6 \text{ Pa}$ C $5.7 \times 10^6 \text{ Pa}$ D $1.3 \times 10^7 \text{ Pa}$